Submission PHYS1-PS2-SUB08

Question PHYS1-PS2-Q01
scan faint / margin cut off
visible work:
$v=v$■$+at=5+3(4)=17$ m/s... ?
last line is hard to read; maybe $v=17$ m/s; displacement=44 m

Question PHYS1-PS2-Q02
scan faint / margin cut off
visible work:
$N=mg=29.4$ N; friction=$\mu N=5.9$ N... ?
last line is hard to read; maybe $a=4.37$ m/s²; friction opposes motion

Question PHYS1-PS2-Q03
scan faint / margin cut off
visible work:
$mgh=\frac{1}{2}mv^2 \Rightarrow v=\sqrt{(2gh)}=\sqrt{(2\cdot 9.8\cdot 6)}$... ?
last line is hard to read; maybe $v=10.84$ m/s

Question PHYS1-PS2-Q04
scan faint / margin cut off
visible work:
p_total,before = p_total,after... ?
last line is hard to read; maybe total momentum before equals after when external impulse is negligible

Question PHYS1-PS2-Q05
scan faint / margin cut off
visible work:
Impulse is area under F(t): $J=F\Delta t=6(0.5)=3$ N·s.... ?
last line is hard to read; maybe impulse=3 N·s; momentum changes by 3 kg·m/s

Question PHYS1-PS2-Q06
scan faint / margin cut off
visible work:
$v=v$■$+at=4+2(5)=14$ m/s... ?
last line is hard to read; maybe $v=14$ m/s; displacement=45 m

Question PHYS1-PS2-Q07
scan faint / margin cut off
visible work:
$N=mg=19.6$ N; friction=$\mu N=3.9$ N... ?
last line is hard to read; maybe $a=3.54$ m/s²; friction opposes motion

Question PHYS1-PS2-Q08

scan faint / margin cut off

visible work:

$mgh=\frac{1}{2}mv^2 \Rightarrow v=\sqrt{(2gh)}=\sqrt{(2\cdot 9.8\cdot 6)}$... ?

last line is hard to read; maybe v=10.84 m/s